Queen Mary
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EBU5376
Digital Signal Processing

Discrete-Time Transformation

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Outline

- Discrete-Time Fourier Transform (DTFT)
- z-Transform
- Linear Convolution using DTFT/z-transform

From CTFT to DTFT

- Previously, we derived the relationship: sampling in time ↔ periodic in frequency.
- This lecture will extend that foundation by deriving the mathematical formula, namely, discrete-time Fourier transform (DTFT), which calculates the frequency spectrum specifically for discrete-time signals.
- DTFT is a special case of CTFT, by viewing the discrete sequence as a continuous function that consists of impulses (e.g., samples)

n-th sample in discrete signal

T_s is the sampling interval

$$x[n] = x(t)\delta(t - nT_s) = x(nT_s)$$

Continuous signal

Hence, we can equivalently express the discrete signal in a continuous form

$$x(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT_s)$$

From CTFT to DTFT

- The formula of CTFT is $X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$
- Here $x(t)$ is a continuous-time signal representing sampled values, with pulses spaced T_s, the sampling interval, apart.

$$X(\omega) = \int_{t=-\infty}^{\infty} \left(\sum_{n=-\infty}^{\infty} x[n]\delta(t - nT_s) \right) e^{-j\omega t} dt$$
$$= \sum_{n=-\infty}^{\infty} x[n] \int_{t=-\infty}^{\infty} e^{-j\omega t} \delta(t - nT_s) dt$$

- The quantity $e^{-j\omega t}\delta(t - nT_s)$ is zero except when $t = nT_s$
- Hence, the integral $\int_{t=-\infty}^{\infty} e^{-j\omega t}\delta(t - nT_s)dt = e^{-j\omega t}$. The CTFT becomes

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n T_s}$$

almost there…

From CTFT to DTFT

- Previously we obtained $X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n T_s}$

- We denote $f_s = \frac{1}{T_s}$ as the sampling rate. Rewriting the expression in terms of the angular frequency, we have

$$= \sum_{n=-\infty}^{\infty} x[n]e^{-j2\pi \frac{f}{f_s} n}$$

$$X(\omega') = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega' n} \quad \Rightarrow \text{ DTFT}$$

- In the above expression, the frequency variable $\omega' = 2\pi \frac{f}{f_s}$ is known as the **normalized frequency**, meaning that the horizontal axis in the frequency domain is now expressed **relative to the sampling frequency** f_s instead of an absolute scale

From CTFT to DTFT

- CTFT: $$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$
- DTFT: $$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- In CTFT, the frequency variable ω has a range of $(-\infty, \infty)$. In DTFT, normalized frequency variable also has the range of $(-\infty, \infty)$.
- But, the output of DTFT must be interpreted relative to the sampling frequency. For example,
  - In CTFT, $\omega = 2\pi$ corresponds to the frequency of 1Hz
  - In DTFT, $\omega = 2\pi$ corresponds to a frequency equal to the sampling frequency
- For DTFT, analyzing the spectrum in $\omega \in (-\pi, \pi)$ provides all the necessary information, do you know why? (Hint: Sampling Theorem)

Properties of DTFT

- Linearity

$$ax_1[n] + bx_2[n] \overset{DTFT}{\longleftrightarrow} aX_1(\omega) + bX_2(\omega)$$

- Time-shift

$$x[n-m] \overset{DTFT}{\longleftrightarrow} e^{-j\omega m}X(\omega)$$

- Time scaling

$$x[an] \overset{DTFT}{\longleftrightarrow} X\left(\frac{\omega}{a}\right)$$

- Frequency-shift

$$e^{j\omega_0 n}x[n] \overset{DTFT}{\longleftrightarrow} X(\omega - \omega_0)$$

- Periodicity

$$X(\omega) = X(\omega + 2\pi k)$$

- Convolution

$$x_1[n] * x_2[n] \overset{DTFT}{\longleftrightarrow} X_1(\omega)X_2(\omega)$$

Inverse CTFT vs. Inverse DTFT

- Inverse CTFT: $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$
- Inverse DTFT: $x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\omega)e^{j\omega n}d\omega$

- The inverse CTFT computes an infinite integral over $(-\infty, \infty)$
- The inverse DTFT computes a finite integral over $(-\pi, \pi)$
  - This simplified equation is due to the periodicity of DTFT
- The formula of inverse DTFT can be derived from inverse CTFT, similar to how the formula of DTFT is derived. Try it yourself. (Hint: $x[n] = x(t)\delta(t - nT_s)$)

Summary – DTFT

DTFT and CTFT have some similarities and differences:

- Time domain
  - Both can be applied to periodic and aperiodic signals
  - Both assumes the input signal has infinite length
  - CTFT requires continuous-time signals, while DTFT works with discrete-time signals

Summary – DTFT

DTFT and CTFT have some similarities and differences:

- Frequency domain
  - Both produce a continuous frequency spectrum
  - DTFT produces a periodic spectrum, while CTFT produces an aperiodic spectrum
  - In DTFT, the frequency variable is normalized to a range of $[-\pi, \pi]$, which corresponds to the Nyquist interval of $\left[-\frac{f_s}{2}, \frac{f_s}{2}\right]$ in terms of the sampling frequency f_s.

Outline

- Discrete-Time Fourier Transform (DTFT)
- z-Transform
  - Relationships with DTFT and Laplace Transform
- Linear Convolution using DTFT/z-transform

Laplace Transform vs. z-Transform

- Previously, we studied the **Laplace transform** as a continuous-time transformation method for solving differential equations and analyzing analog systems.
- For digital signals, the equivalent method to Laplace transform is called the z-transform
- **Z-transform** is utilized for solving difference equations and analyzing digital systems

$$\frac{d}{dt}y(t) = x(t)$$

A differential equation

$$y[n] - y[n-1] = x[n]$$

A difference equation

Z-transform

- The z-transform of a sequence $x[n]$ is given by

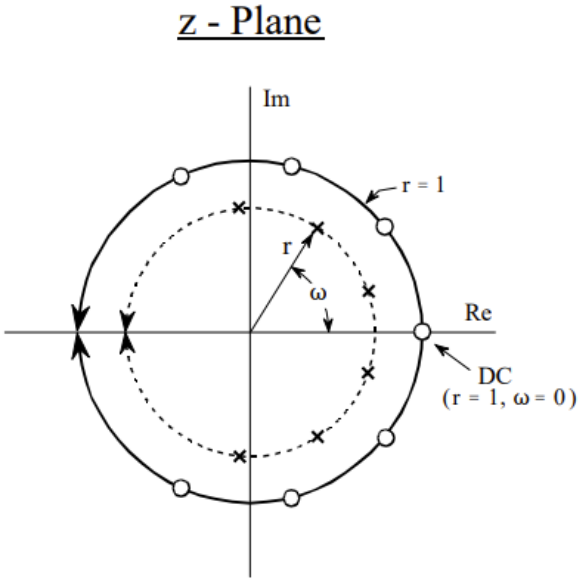
$$X(z) = Z(x[n]) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Equivalently, we have

$$X(z) = \cdots + x[-1]z + x[0]z^{-0} + x[1]z^{-1} + x[2]z^{-2} + \cdots$$

- Here, $z = re^{j\omega} = r(\cos(\omega) + jsin(\omega))$ is a complex variable in the z-plane with $r > 0, j = \sqrt{-1}$
  - r is the magnitude of z, i.e., $r = |z|$
  - ω is the phase/angle of z
  - So, z-transform can also be written as

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]r^{-n}e^{-j\omega n}$$



Z-transform

- **Examples:**

1. The z-transform of $x[n] = [1, 2, 3, 4]$ is given by

2. The z-transform of $x[n] = [1, 0, 0, 1]$ is given by

3. The z-transform of $x[n] = u[n]$ is given by

- For finite sequences, when not specified, we assume that the first value in the sequence is indexed with $n = 0$

Table of Common z-transform Pairs

Time domain	z-domain
$\delta[n]$	1
$u[n]$	$\frac{z}{z-1}$
$a^n u[n]$	$\frac{z}{z-a}$
$na^n u[n]$	$\frac{az}{(z-a)^2}$

- This table assumes right-sided sequences in the time domain ($n \geq 0$) as they are practical for real-world DSP.
- Other sequence types (e.g. left-sided and two-sided) are mainly theoretical and less relevant in practice.

Inverse z-transform

- Given the z-transform $X(z) = Z(x[n])$, the inverse z-transform is formulated as

$$x[n] = \frac{1}{2\pi j}\oint_C X(z)z^{n-1}dz$$

  - C is a closed contour in the region of convergence of $X(z)$
  - However, direct evaluation of this integral is rarely practical, so alternative methods are used.

- There are many methods to obtain the inverse z-transform. In this module, we will focus on the **partial fraction expansion method:**
  - For simple Z-transform functions, use the Table of common z-transforms pairs to directly determine the inverse.
  - For complex functions, decompose them into partial fractions and apply the linearity property with the Table.

Inverse z-transform

- <u>Example</u>: Compute the inverse z-transform of $X(z) = \frac{z}{z-1} + \frac{z}{z-3}$

Outline

- Discrete-Time Fourier Transform (DTFT)
- z-Transform
  - Forward z-transform
  - Inverse z-transform
  - Relationships with other transform methods
- Linear Convolution using DTFT/z-transform

Relating z-Transform and DTFT

- **DTFT** $$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$
- **z-transform** $$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \cdot r^{-n}$$

- The parameter of DTFT is ω, a real-valued parameter
- The parameter of z-transform is $z = re^{j\omega}$, a complex-valued parameter that consists of two real-valued hyperparameters, r and ω
- When $z = e^{j\omega}$, the result of z-transform is identical to the result of DTFT.
- In other words, the values on the unit circle $|z| = 1$ in the z-plane are the values of the DTFT of that signal.

Relating z-Transform and DTFT

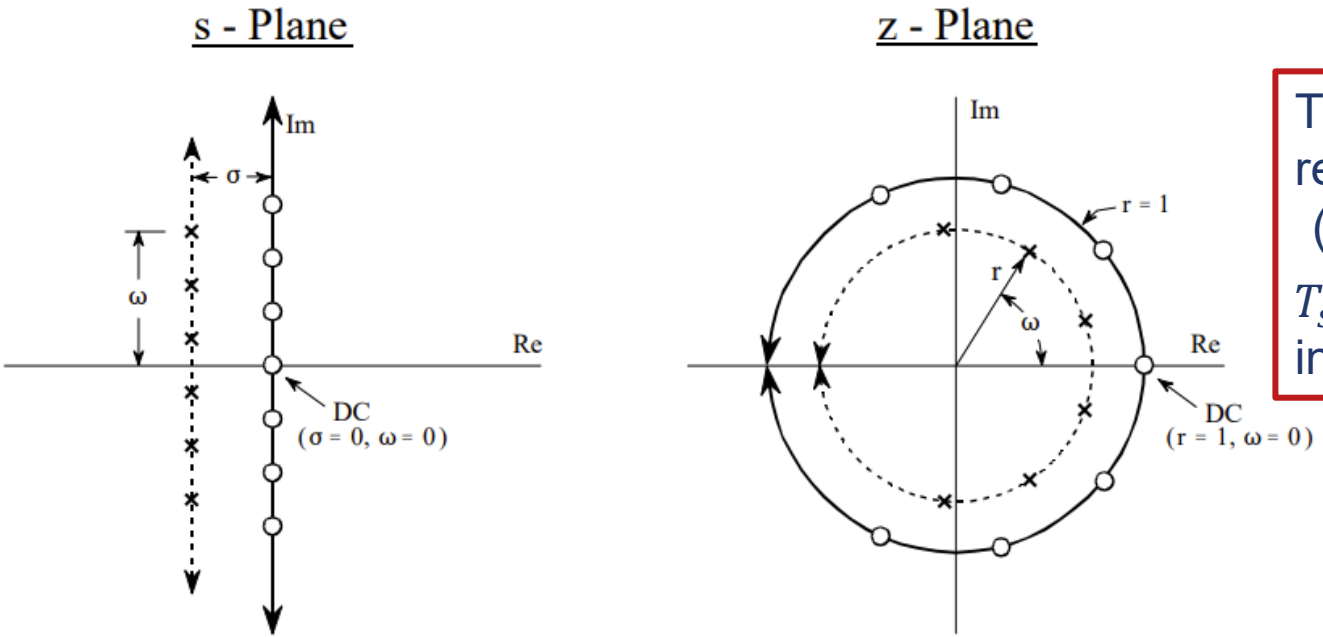
- **DTFT** analyzes how different frequency components contribute to a signal and is used for frequency analysis. It corresponds to the z-transform on the unit circle ($r = 1$), meaning it only considers steady-state oscillations
- The **z-transform** extends analysis beyond the unit circle by introducing r as a variable, allowing examination of signal behavior in terms of growth or decay. This shift from purely oscillatory analysis to a more general representation enables system analysis, capturing not just frequency content but also how signals behave over time, including growth and decay

Relating z-Transform and Laplace Transform

- There are two types of signal processing systems: **analog and digital**.
  - **Analog systems** operate on continuous-time signals, which require the Laplace transform for analysis.
  - **Digital systems** operate on discrete-time signals, and their behavior is analyzed using the z-transform (as introduced earlier).
- The z-transform is the discrete counterpart of the Laplace transform.
- **A digital system can be seen as a sampled version of an analog system**.
  - A continuous-time system, when sampled at intervals of T_s, becomes a discrete-time system.
  - Since the discrete-time domain originates from sampling the continuous-time domain, we can establish a mathematical relationship between the two.

Relating z-Transform and Laplace Transform

- For **Laplace transform** ($s = \sigma + j\omega$): The s-plane is a rectangular coordinate system with ω expressing the distance along the real (horizontal) axis, and σ the distance along the imaginary (vertical) axis.
- For **z-transform** ($z = re^{j\omega}$): The z-plane is in polar form, with r being the distance to the origin, and ω the angle measured to the positive horizontal axis.

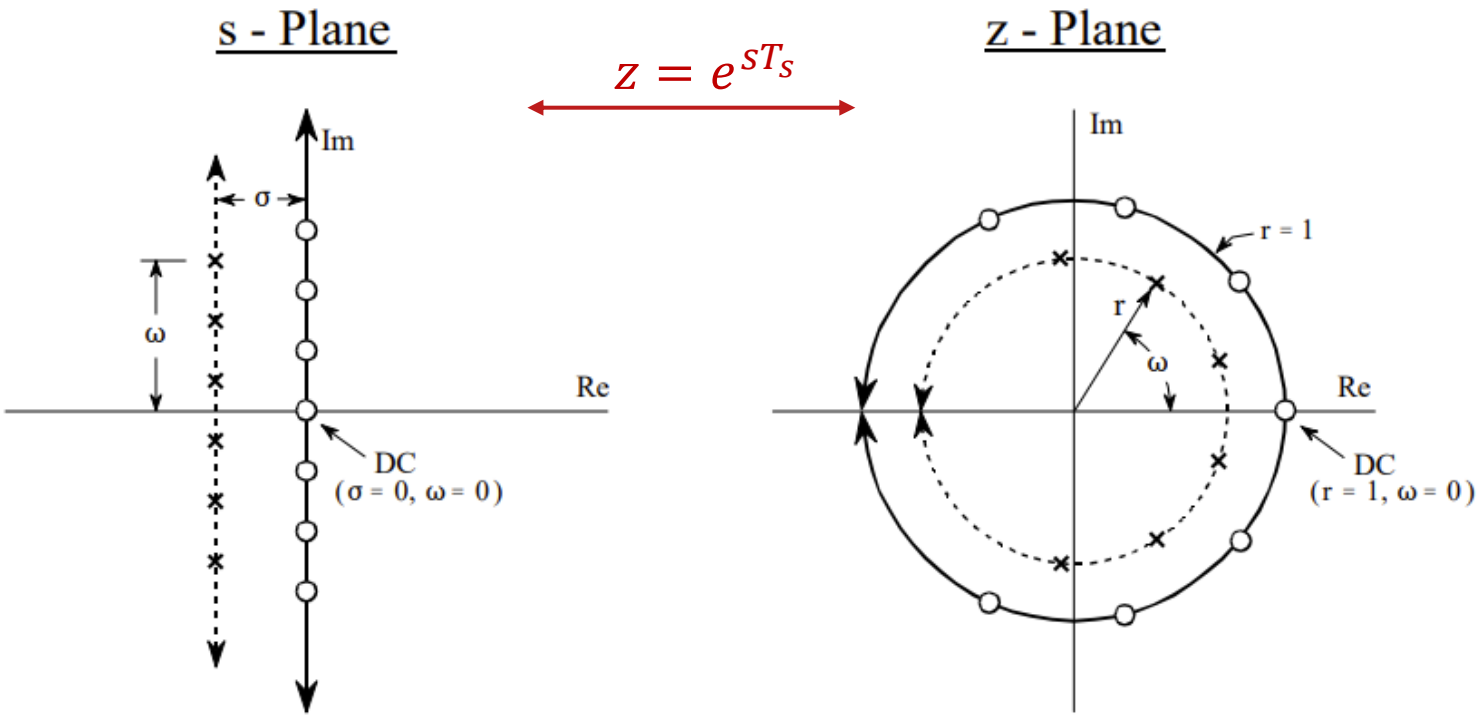


The two planes are related by $z = e^{sT_s}$ (Proof in Appendix A)

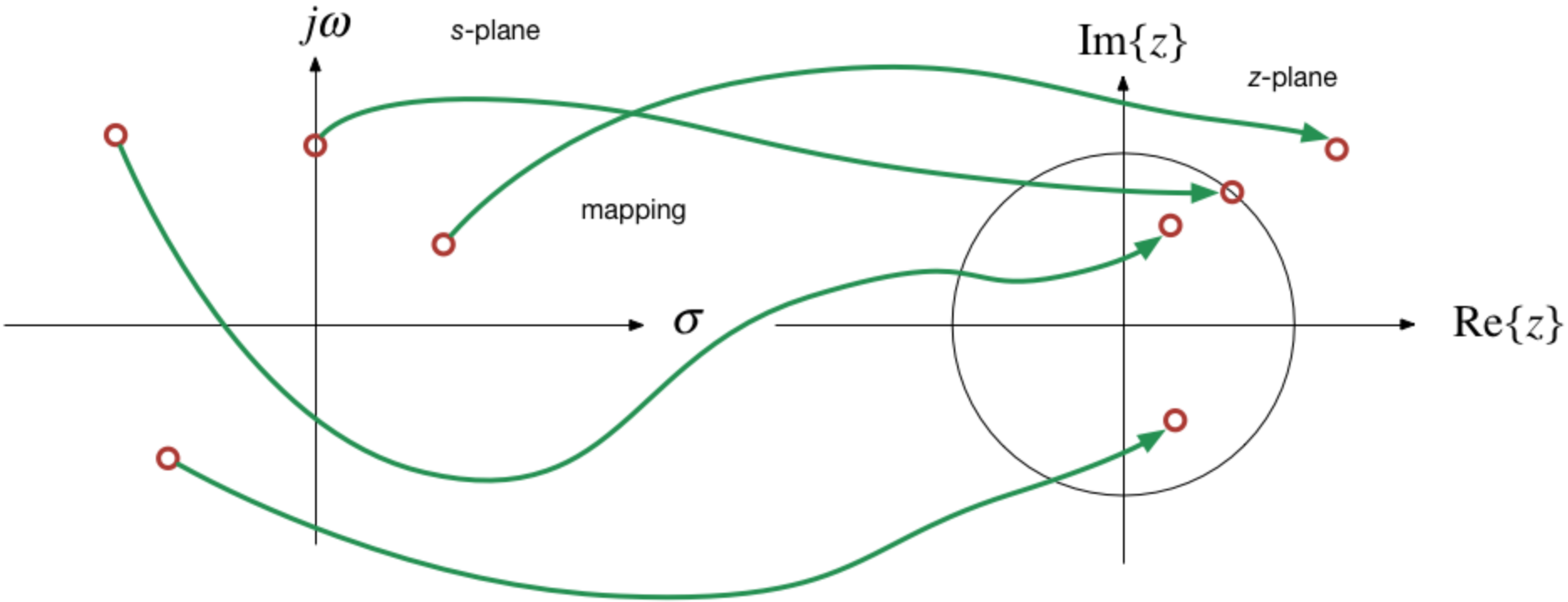
T_s is the sampling rate in the time-domain

s-domain vs. z-domain

- Values along a vertical line in the s-plane ($s = j\omega$) correspond to values along a circle in the z-plane ($z = e^{j\omega T_s}$).



s-plane vs. z-plane



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The discrete convolution theorem - DTFT

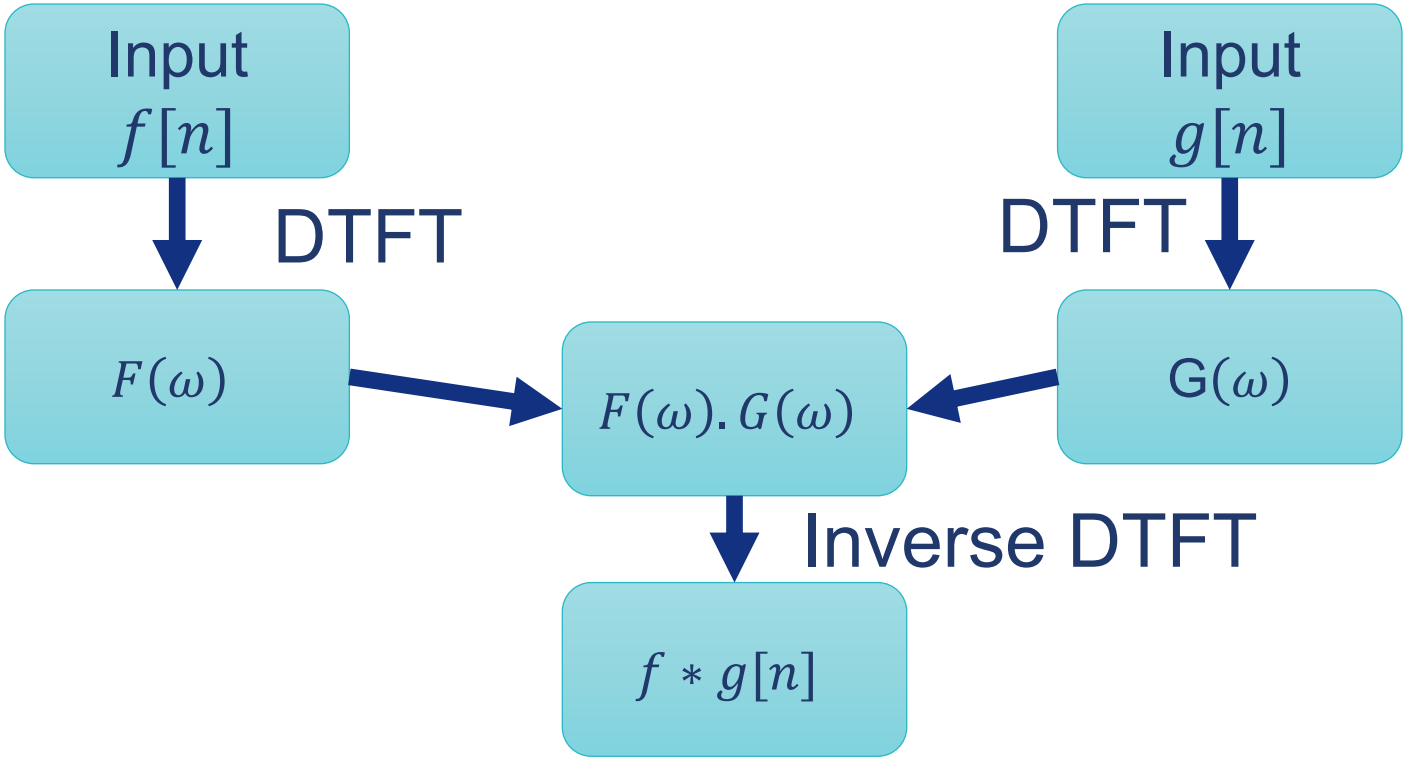
- The convolution theorem also holds for discrete systems with infinite sequences $f[n]$ $g[n]$ and their DTFTs $G(\omega)$ $F(\omega)$

$$\mathcal{F}\left((f * g)[n]\right) = F(\omega)G(\omega)$$

$$(f * g)[n] = \mathcal{F}^{-1}\left(F(\omega)G(\omega)\right)$$

- Where $\mathcal{F}$ and $\mathcal{F}^{-1}$ now represent the DTFT and Inverse DTFT

Linear Convolution via DTFT/z-transform



Linear Convolution via DTFT/z-transform

- Two sequences: $x[n] = [2, 2, 2]$ and $h[n] = [1, 1, 1, 1]$
- Compute the linear convolution $x[n] * h[n]$ using DTFT or z-transform.

From Infinite-length to Finite-length Signals

- **DTFT is typically defined for sequences that have infinite length**
- Real-world signals have finite length
- **If $\boldsymbol{x[n]}$ is truly infinite in length**, the DTFT exactly describes the signal's frequency content
- **If $\boldsymbol{x[n]}$ is finite** (defined for $0 \leq n < N$), then we assume $x[n] = 0$ for all other n
  - The DTFT formula still applies, but the finite-length nature of the signal introduces side effects, such as spectral leakage
  - The DTFT behaves as if the original infinite-length signal was truncated
- In DTFT, there are **no "purely finite" signals**—every finite signal must be considered a special case of an infinite signal with zeros extending to infinity.

From Infinite-length to Finite-length Signals

- What happens to the frequency spectrum, when DTFT is applied to a finite-length signal? How to find out?
- In the time-domain, the truncation of an infinite-length signal $x[n]$ into a finite-length signal can be mathematically modelled as multiplying $x[n]$ with a rectangular sequence:

$$x_{win}[n] = x[n] \cdot w_N[n]$$

Where

- $x_{win}[n]$ is a signal of length N
- $w_N[n]$ is a rectangular window defined as

$$w_N[n] = \begin{cases} 1, & 0 \leq n < N \\ 0, & otherwise \end{cases}$$

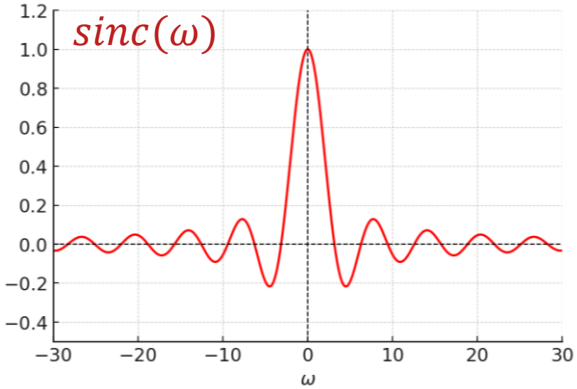
- By the convolution theorem, this means the spectrum of $x[n]$ is convolved with the spectrum of $w_N[n]$

From Infinite-length to Finite-length Signals

- Since **multiplication in the time domain** corresponds to **convolution in the frequency domain**, the DTFT of $x_{win}[n]$ is:

$$X_{win}(\omega) = X(\omega) * W_N(\omega)$$

  Where $W_N(\omega)$ is the DTFT of the rectangular window $w_N[n]$, which is known to be a sinc function



- How this leads to spectral leakage:
  1. Sinc function has side lobes that extend **infinitely**
  2. Convolving with $W_N(\omega)$ introduces side lobes into the spectrum, distributing $X(\omega)$ into neighboring frequencies
  3. This results in frequency components extending beyond their original range

- **The wider the time window, the narrower the sinc**: As N increases, $W_N(\omega)$ becomes narrower, and the spectral leakage reduces

Summary

- So far, we have studied the time-frequency transformations of both continuous-time signals and discrete-time signals:

  Discrete time Fourier transform (DTFT): $X(\omega) = \sum_{-\infty}^{\infty} x[t]e^{-j\omega t}$

Continuous time Fourier transform (CTFT): $X(f) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$

- DTFT can be derived from z-transform ($z = re^{j\omega}, r = 1$), where CTFT can be derived from Laplace transform ($s = \sigma + j\omega, \sigma = 0$)
- DTFT is the discrete equivalent of CTFT, similarly, the z-transform corresponds to the discrete version of the Laplace transform.

Summary

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- DTFT can be derived from z-transform ($z = re^{j\omega}, r = 1$), where CTFT can be derived from Laplace transform ($s = \sigma + j\omega, \sigma = 0$)
- DTFT is the discrete equivalent of CTFT, similarly, the z-transform corresponds to the discrete version of the Laplace transform.
- The DTFT defines the frequency content of a signal but does not give a discrete set of values that can be computed in practice.
- To analyze or process signals on a computer, we need a finite number of frequency points that still retain key spectral information.

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Appendix A: Relating Laplace transform and z-transform

- We can derive the z-transform using the equation of the Laplace transform:

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt\,, \qquad s = \sigma + j\omega$$

Laplace transform: $\Rightarrow X(\sigma,\omega) = \int_{-\infty}^{\infty} x(t)e^{-\sigma t}e^{-j\omega t}\,dt$

Discretize:
$x[n] := x(nT_s)$

$$\Rightarrow X(\sigma,\omega) = T_s \sum_{n=-\infty}^{\infty} x[n]e^{-\sigma n T_s}e^{-j\omega n T_s}$$

$$\Rightarrow X(\sigma,\omega) = T_s \sum_{n=-\infty}^{\infty} x[n]e^{-nT_s(\sigma+\omega j)}$$

z-transform: $\Rightarrow X(z) = T_s \sum_{n=-\infty}^{\infty} x[n]z^{-n}\,, \qquad z = e^{T_s(\sigma+\omega j)} = e^{sT_s}$